

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

EE 19-209-0403 ELECTRICAL MACHINES II

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: To summarise the construction and principle of operation of synchronous machines and induction machines.

CO2: To analyze the parallel operation of alternators and the performance of synchronous motors.

CO3: To determine the voltage regulation of alternators.

CO4: To identify starting techniques of three phase induction motor.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a) Derive the relation between the number of poles, frequency and speed of the alternator.	3	L2	1	1	
	(b) An alternator has 18 slots/pole and the first coil lies in slots 1 and 16. Calculate pitch factor for (i) fundamental (ii) 5 th harmonic (iii) 7 th harmonic.	3	L3	1	1,2,3	
	(c) What are the conditions to be fulfilled for parallel operation of two alternators?	3	L1	2	1	
	(d) Explain what do you understand by hunting in a synchronous machine.	3	L2	2	1	
	(e) A 3 phase induction motor has 2 poles and is connected to 400 V, 50 HZ supply. Calculate the actual rotor speed and rotor frequency when the slip is 4%.	3	L3	1	2,3,4	
	(f) Derive the expression for torque developed in a 3 phase induction motor.	3	L2	4	1,2	
	(g) Draw the torque slip characteristics of a single phase induction motor.	3	L2	4	1,2	
	(h) Explain the working principle of a shaded pole motor.	3	L2	1	1	

PART B

(4 × 12 = 48)

II.	(a) Compare salient pole and non-salient pole synchronous machines.	6	L2	1	1
	(b) The stator of 3 phase, 20 pole alternator has 120 slots and there are 4 conductors per slot. If the speed for the alternator is 300 rpm, calculate the induced emf per phase. The flux per pole is 0.05 wb. Assume coil span as 160°.	6	L3	1	1,2,3

OR

III.	A 3.3 KV, 3 phase star connected alternator has a full load current of 100 A. Under short-circuit condition it takes 5 A field current to produce full load short circuit current. The emf on open circuit for the same field current is 900 V (line to line). The armature resistance is 0.9 Ω/ phase. Determine the voltage regulation at full load, 0.8 pf lagging and 0.8 pf leading.	12	L3	3	2,3,4
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(P.T.O.)

		Marks	BL	CO	PO								
IV.	(a) Explain with diagram any one method of synchronizing 2 three phase alternators.	6	L2	2	1,2								
	(b) Two ac generators running in parallel supply a lightening load of 200 kW and a motor load of 400 kW at 0.8 pf lagging. One machine is loaded 2400 kW at 0.95 pf lagging. What is the output and power factor of the second machine?	6	L3	2	2,3,4								
OR													
V.	(a) Explain the effects of varying the excitation on armature current and power factor of a synchronous motor with relevant vector diagrams.	6	L2	2	1,2								
	(b) A 2.3 kV, 3 phase star connected synchronous motor has $Z_s = (0.2 + j 2.2) \Omega$ /phase. The motor is operating at 0.5 power factor leading with a line current of 200 A.	6	L3	2	2,3,4								
VI.	(a) The rotor resistance and reactance per phase of a 4 pole, 50 Hz, 3 phase induction motor are 0.25 and 0.12 Ω respectively. (i) Find the speed at maximum torque (ii) Find the value of additional rotor resistance per phase required to give three fourth of maximum torque at starting.	6	L3	4	1,2,3								
	(b) Describe the no load and blocked rotor test on an induction motor.	6	L2	1	1,2								
OR													
VII.	A 3 phase, 400 V induction motor gave the following test reading:	12	L3	1	1,2,3,4								
<table border="1"> <tr> <td>No load</td> <td>400 V</td> <td>1250 W</td> <td>9 A</td> </tr> <tr> <td>Blocked rotor</td> <td>150 V</td> <td>4 kW</td> <td>38 A</td> </tr> </table>		No load	400 V	1250 W	9 A	Blocked rotor	150 V	4 kW	38 A				
No load	400 V	1250 W	9 A										
Blocked rotor	150 V	4 kW	38 A										
Draw the circle diagram. If the normal rating is 14.9 kW, find from circle diagram, full load value of current, power factor and slip.													
VIII.	Explain double cage induction motor with relevant diagrams. Draw the equivalent circuit and torque slip characteristics.	12	L2	1	1,2								
OR													
IX.	(a) Explain how the speed of slip ring induction motor can be changed by changing the rotor circuit resistance.	6	L2	1	1,2								
	(b) Draw and explain capacitor start induction motor.	6	L1	1	1								

Blooms's Taxonomy Level's

L1 - 7.5%, L2 - 47.5%, L3 - 45%.
